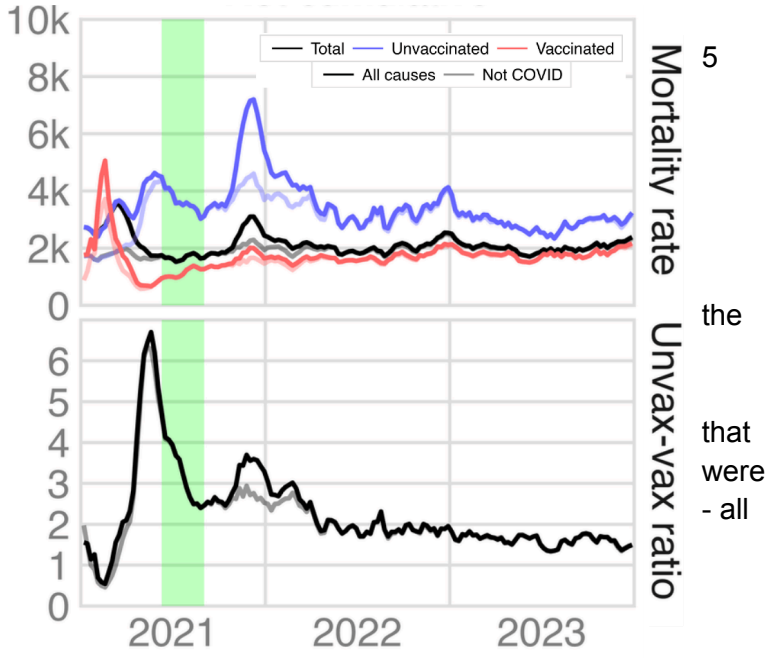
Kirsch's concession: Hybrid VE=83%

Kirsch has conceded, based on CZ data, that the vaccines saved lives (key point 8). This is indeed hard to deny given the sharp increase in unvaccinated ACM during covid waves. He estimates a reduction in covid deaths of 83%. While he only mentions it in the context of 2 months of peak covid, the vaccine does not lose effectiveness when infections decline, therefore this VE applies to the entire period.
Note that 83% is the actual empirical reduction, including NI (i.e. hybrid VE). It is therefore compatible with all other studies of VE=90%, and should also imply Kirsch agrees with our lives saved estimate of 1,174,000 lives, since that is calculated from VE=90%.

Kirsch's updated claim is that there is still net harm because VIDs are higher than lives saved. This means he's claiming over 1,174,000 VIDs, all of which must be NCVIDs (as he's abandoning his claim of vaccines increasing CFR and cases).
This is in stark contradiction to the evidence, which shows that even 0.1% that number is impossible, as we detailed here. Specifically: a) the complete lack of EDs in vaccinated zero covid countries, b) the lack of EDs in highly-vaccinated US states during covid troughs, c) the lack of EDs that cannot be attributed to known causes, and d) the "failure" of any health authority anywhere in the world to detect more than a handful of NCVIDs.
Note that Kirsch doesn't provide an estimate of VIDs in his new model, but claimed 415K in V3 (only 266K of which were NCVIDs) and 497K in V4 (400K NCVIDs). Thus, even according to Kirsch's impossible figures, vaccines saved many more than they killed.

This alone is enough to confidently determine "net benefit" is true. The rest of this document will just provide rebuttals to specific claims made in Kirsch's pages.



KCOR

Kirsch is claiming his proprietary method shows net harm, which as explained above, means an impossible ~1.2M NCVIDs. So what is the source of error?
Instead of calculating deaths and lives saved separately, he proposes a new model called KCOR tracks cumulative deaths over time in cohorts who vaccinated and unvaccinated at a certain cutoff date relative to their deaths at a baseline period with low covid.
This gives him an overall higher death ratio in the vaccinated, despite an advantage to vaccinated during covid waves.
There are many mistakes in this model, but the main one is enough to understand why it generated such a large error. It is the same mistake he made in his previous CZ analysis, which we explain here. When looking at deaths over time, it is clear that vaccinated always die less. This is a combination of two effects: HVE and VE.
In an attempt to control for HVE, Kirsch chooses a low covid baseline period that is immediately after vaccinations (marked green). While such a period does neutralize VE (there are no Covid deaths to save), it can't be used as an HVE baseline, since HVE is not constant - it gradually declines over several months (bottom chart). By choosing a high-HVE period as baseline, he is wrongly interpreting the decline in HVE as vaccinated deaths.

Conversely, in our analysis, we track HVE over time, and dynamically adjust for it, rather than dividing by a constant HVE. That method produces similar numbers to all other models: VE=90% and 700,000 lives saved.

Levi et al

This **preprint** claims a 36% higher NCACM for pfizer over moderna. The first glaring problem is that it is the exact opposite of Kirsch's previous claim of 30% higher mortality in Moderna (main doc p. 28), which he attributed to its higher dose. We showed that his analysis did not control well for comorbidities. The new study seems to have the same problem in that comorbidities are not in the matching criteria (as recognized by the authors in the limitations section).
Interestingly, Levi et al reports that the difference in deaths between brands was highest for Covid deaths. That is strong evidence that the difference is due to comorbidities (as Covid deaths are highly correlated with comorbidities).
Note that since the parties now agree that the vaccines reduce Covid deaths, we can see that the difference in brands can occur without it being indicative of VID, invalidating the whole study.
The most reasonable explanation is that neither brand is more dangerous, and they are simply given to populations with different risk profiles in different geographies, due to supply issues, doctors' decisions, insurance agreements etc.
There are many ways to assess vaccine safety that are far more accurate than comparing brands, and we have shown all of them point to negligible VIDs.

Response to Evidence List

1. Levi et al discussed above
2. Time series - On short time scales, vaccinated deaths are artificially lower due to HVE. Their return to normal should not be interpreted as harm. Kirsch compares it to HVE in flu vaccines, but each vaccine is administered differently so has different HVE. We have real examples of Covid HVE in the CZ, ONS, and NL ACM data we analyze in V5, and they show a sharp decrease in the first weeks.
   As to Israel's wastewater data - It simply shows the obvious finding that many people were infected during omicron and delta. This says nothing on how many would have died absent the vaccines.
3. Aarstad ONS - Based on the same mistake in interpreting HVE that Kirsch did in KCOR.
4. Morstert - The EDs are covid deaths, as evident by comparing each country in figure 2 to its covid deaths chart. The lack of EDs during covid troughs (especially evident during Australia's zero covid period) indicates these are not VIDs.
5. Aarstad EU - This is based on a faulty extrapolation of expected deaths. Once corrected, vaccination rates are associated with lower deaths. Detailed response here
6. "Paradoxical" - Comparison of countries is highly confounded. As with the Subramanian letter we debunked in V4, the correlation is likely due to poorer countries having lower vaccination but also a younger population and underreported covid deaths.
7. The Norway paper discusses "very frail people with very short life expectancy" in nursing homes, so whether or not it found real VIDs, this has little relevance to the debate question. We are willing to concede that a treatment that weakens some healthy people for a few days, can accelerate the death of someone nearing death.
8. Redert - Low quality study based on an error-ridden model that produces impossible VID estimates that far exceed total EDs . Detailed review here.
9. The parties agreed that the documents be readable without following links. Placing a github link is not evidence, and we cannot afford to debug yet another model that didn't pass peer-review. In any case, we fully analyzed the CZ data and showed it strongly supports the net benefit hypothesis.
10. Same response as 9.

11. The Palinkas study actually showed vaccinated fared relatively better during covid waves - similar to our findings on the CZ data. The figures Kirsch cherry-picks from the study are not adjusted, and once corrected, show vaccine benefit: "...survival probabilities were better in the unvaccinated… but the multivariate models showed better survival for each vaccination group".
12. The parties agreed that the documents be readable without following links. This provides no details on the autopsies. A quick review of Kirsch's blog post shows three of the five received Janssen or AstraZeneca (non mRNA), both known to cause rare deaths, and both rarely used in the US. This, therefore, actually supports the safety of mRNA vaccines. In any case, having five VIDs is consistent with our hypothesis.
13. Kirsch relies heavily on ChatGPT for evaluating evidence. We note that LLM sycophancy, the "propensity of models to excessively agree with or flatter users, often at the expense of factual accuracy" is a known problem, which we could easily use to create the opposite responses. Unless instructed otherwise by the judges, we are ignoring these.

Answers to Questions

1. Our models are fully disclosed, including links to spreadsheets. Unclear what validation is required, but we've shown the lives saved calculated by our models using VE are equal to those calculated from CZ, US ACM, and deaths by age.
2. Discussed above.
3. CZ data is good and shows VE=90%, and the same lives saved as other models. We're not aware of better record-level datasets, but we recommend against relying on a single dataset. Our case is built from a consensus of many parallel data sources and methodologies, where each one has different weaknesses, creating a highly robust conclusion.
4. The data seems valid. It is only the KCOR model that is invalid.
5. Discussed above. Note that Kirsch's position is that CZ data is the most reliable, and in his previous work he claimed it showed Moderna is more lethal - contradicting Levi et al.
6. See 3.
7. Yes. Discussed above.
8. The mistake was discussed above (using a high HVE period as baseline). When charting vaccinated and unvaccinated deaths over time (not cumulative!) HVE and VE are easy to view separately, and it is clear the vaccine saves lives.
9. In the US these are mostly Covid deaths. We did a detailed breakdown of ED causes in the US. There is little value in repeating this for these countries.
10. No one calculates EDs this way. Specifically, the mistake here is forgetting that deaths are on a constant rise due to a growing elderly population. We've done detailed ED analysis using proper methodologies.
11. 2021 ED wasn't higher when considering 2020 was 9 months of Covid. 2022 ED was the lowest. In any case, there were far more infections in both years and without vaccines, ED would have been far higher.
12. Eyeballing cumulative deaths isn't how you assess vaccine efficacy. In any case, deaths continued due to more covid infections. Without vaccines, deaths would have been far higher than shown in the chart, as seen in the comparison of Hong Kong (with low vaccination rate in elderly) to the other zero covid countries.

Summary

This goes back to our argument about inconsistencies in the net harm hypothesis. Kirsch has abandoned most of his previous claims, and now relies on a completely new model, while not realizing he's claiming an incredibly impossible number of VIDs.

If there was a good case to make for net harm, one of the thousands of people who are working relentlessly to find it would have produced it by now, and Kirsch would present it. There is none because it's impossible to find good evidence for a hypothesis that is 1000x off the mark.